Vacuum 220 (2024) 112769

Contents lists available at ScienceDirect

Vacuum

journal homepage: www.elsevier.com/locate/vacuum

# Additive manufacturing for ultra-high vacuum components: Leveraging photo-polymer resin technologies

Jiaqi Li [a,*], Thomas Mcpartland [b], Bernardo Gutierrez [b], Jan Pedersen [a], Yan Zhou [b]

[a] Department of Computer Science, University of Nevada, Las Vegas, 89154, Nevada, USA
[b] Department of Physics and Astronomy, University of Nevada, Las Vegas, 89154, Nevada, USA

ARTICLE INFO

Handling Editor: Prof. L.G. Hultman

*Keywords:*
Ultra-high vacuum
Additive manufacturing
Photo-polymer resin
Vacseal coating

ABSTRACT

Over the past two decades, Additive Manufacturing (AM) technologies, commonly known as 3D printing, have witnessed exponential advancements, transforming the paradigms of instrument design and fabrication across an array of scientific and engineering disciplines. The utilization of Computer-Aided Design (CAD) and monolithic machining within the AM framework offers unprecedented levels of flexibility, precision, and mechanical stability. These attributes not only facilitate complex geometrical constructions but also substantially streamline the overall fabrication workflows. In this study, we provide experimental evidence to substantiate the viability of employing Masked Stereolithography Additive Manufacturing (MSLA) technology in concert with commercially available photo-polymer resins for the fabrication of components that are compatible with Ultra-High Vacuum (UHV) conditions. Utilizing high-temperature photo-polymer resins and specialized post-printing treatments—including an approximate 24-h vacuum bake—we have achieved a vacuum level of approximately $1 \times 10^{-9}$ torr for a solid component encompassing a volume up to 16 $cm^3$. This methodology paves the way for the expedited and cost-effective production of complex, UHV-compatible devices, thereby offering a substantial departure from conventional manufacturing techniques. The wide-ranging applicability of this technology holds considerable potential for transformative impacts across diverse sectors of research and development. By leveraging UHV-compatible MSLA technology, both academic researchers and industrial professionals are poised to benefit from an efficient and adaptable fabrication methodology, consequently catalyzing cutting-edge advancements in their respective fields of inquiry.

## 1. Introduction

Additive Manufacturing (AM), colloquially referred to as 3D printing, is a process of constructing complex geometrical forms through the layer-by-layer deposition of materials. This technique stands in contrast to Subtractive Manufacturing (SM), wherein an object is formed by excising material from a pre-existing solid substrate. Originating in the 1980s as a rapid prototyping modality [1], AM has undergone significant evolutionary strides over the past two decades, transcending its initial application to emerge as a viable pathway for the production of fully functional devices [2–5]. This technological development has engendered a transformative shift in the manufacturing market, offering unparalleled levels of flexibility, adaptability, and innovative capability. This is particularly synergistic with contemporary Computer-Aided Design (CAD) platforms, which inherently benefit from the digital-centric ethos imbued within AM processes [6–9]. Consequently, the industry has witnessed a paradigmatic migration towards design-centric manufacturing methodologies, thereby facilitating the realization of intricate geometrical constructs hitherto considered unattainable [10–13].

Beyond its design-centric virtues, AM confers a multitude of additional merits. Notably, it substantially mitigates production expenditures while concurrently accelerating developmental timelines, thus serving as an enabling framework for bespoke innovation [14]. Furthermore, the method manifests environmental sustainability through a marked diminution of material waste relative to subtractive paradigms, thus aligning with eco-conscious manufacturing imperatives and advancing the cause of responsible resource stewardship [15,16]. The recent endorsement and deployment of AM by industrial stalwarts such as Boeing (Boeing Advanced Research Center at University of Washington) and General Electric [17] underscore its burgeoning significance within the contemporary manufacturing landscape. Moreover,

* Corresponding author.
*E-mail address:* lij50@unlv.nevada.edu (J. Li).

https://doi.org/10.1016/j.vacuum.2023.112769
Received 12 May 2023; Received in revised form 20 September 2023; Accepted 6 November 2023
Available online 17 November 2023
0042-207X/© 2023 Elsevier Ltd. All rights reserved.

applications employing customized materials are proliferating across diverse sectors, including but not limited to, aerospace [18], automotive [19], and medical devices [20]. Such developments further accentuate the versatile and transformative potency of AM as a disruptive technological force within modern industrial ecosystems.

While AM has fundamentally revolutionized diverse sectors of industry, its integration into the fabrication of component for Ultra-High Vacuum (UHV) environments remains a significant challenge, often considered unfeasible. The stringent criteria for UHV-compatibility – ranging from exceedingly low outgassing rates to specialized alloy compositions and ultra-fine surface finishes – have served as considerable impediments to the utilization of AM techniques in this field [18]. Such limitations have consequently restrained the exploitation of AM's inherent advantages in design flexibility and bespoke customization, thereby curtailing its wider adoption within UHV applications. Over the past half-decade, substantial research efforts have been diligently focused on overcoming these obstacles, with the aim of expanding the applicability of AM technologies in the realm of UHV-specific implementations.

Two predominant AM technologies have emerged at the vanguard of this research initiative: Fused Filament Fabrication (FFF) and Direct Metal Laser Sintering (DMLS). Recent technological advancements in FFF, which employs Polylactic Acid (PLA) filaments, have achieved vacuum pressures as low as $1 \times 10^{-6}$ torr [21,22]. Additional refinements have been attained through specialized surface treatments, such as atomic layer deposition of aluminum oxide and post-processing heat treatments, leading to vacuum pressures approaching $1 \times 10^{-7}$ torr [23,24]. On the other hand, DMLS technology has shown extraordinary promise, attaining to vacuum pressures as diminutive as $1 \times 10^{-10}$ torr [25,26]. However, the extensive capital investment and infrastructural prerequisites associated with DMLS render it economically unfeasible for smaller research groups.

In the present study, we demonstrate an advancement in the fabrication of UHV-compatible components through the utilization of Masked Stereolithography Additive Manufacturing (MSLA) technology. This methodology leverages readily available and cost-effective photo-polymer resins, thereby rendering it an eminently pragmatic approach for widespread applications. Our empirical investigation primarily focuses on the systematic optimization of various post-printing procedures, including cleaning, post-curing, surface coating, and high-temperature baking, each specifically designed to enhance vacuum compatibility. Additionally, we scrutinize the performance of three distinct photo-polymer resins – ABS-like, transparent, and high-temperature resins – all endowed with unique physical and chemical properties. Through the implementation of these optimized post-printing processes on the high-temperature resin, we have accomplished a base pressure of approximately $1 \times 10^{-9}$ torr after a 30-h post-printing treatment. This achievement represents an enhancement of two orders of magnitude in comparison to prior investigations that utilize non-metallic materials [21–23,27]. Additionally, we conduct a preliminary examination of the resin's outgassing properties utilizing a Residual Gas Analyzer (RGA). Our finding corroborate that the outgassing of both water and hydrocarbon molecules can be appreciably attenuated via the employment of Vacseal leak sealant, a silicone-based liquid resin amenable to high-temperature curing. Proper surface coating with this Vacseal sealant would not only augments the attainable vacuum level – approaching $1 \times 10^{-10}$ torr – upon extended baking durations (eg., up to one week), but it also drastically minimizes the risk for contaminating the vacuum apparatus via the outgassing of hydrocarbon molecules.

The present methodology demonstrably attains vacuum performance that are commensurate with those achieved via the DMLS technique, albeit at a markedly diminished cost. This synergetic alignment of cost-efficiency and straightforward operational implementation designates our approach as a propitious contender for widespread assimilation into the AM of UHV-compatible components. Furthermore, the conceptual fusion of AM techniques – renowned for their proficiency in managing intricate geometries – with subtractive manufacturing methodologies, distinguished by their capacity to proffer unique material attributes, engenders a novel manufacturing paradigm. This paradigm is capable of concurrently realizing complex structural architectures, generative design principles, and distinct mechanical characteristics. Such a transformative confluence of capabilities portends significant ramifications for both the academic research and industrial manufacturing milieus, serving to broaden the spectrum of feasible AM applications within the domain of UHV technologies. Moreover, it offers the potential to render high-caliber, customized components increasingly attainable for a diverse array of research practitioners and industrial entities.

## 2. Method

In the present study, we assess the vacuum compatibility of three distinct, commercially available photo-polymer resins, as enumerated in Table 1. Each selected resins is economically viable, with a cost not exceeding $100 per kilogram. The resins are chosen based on the following criteria: (1) The ABS-like resin offers superior dimensional stability by effectively mitigating volumetric shrinkage during the curing phase; (2) The transparent resin affords the unique capability of fabricating optically transparent components; and (3) The high-temperature resin demonstrates structural integrity up to a temperature of 220 °C, making it crucial for attaining the targeted UHV conditions. The thermal resistance temperatures of the selected resins, denoted as $T_1$ in Table 1, are sourced from the manufacturers' product specifications. It is imperative to note that diverse manufacturing standards can influence these thermal resistance figures. To ascertain the appropriate baking temperatures for our experimental design, we conduct a series of thermal tests to delineate two critical temperature thresholds: (1) A structural failure temperature, denoted as $T_2$ in Table 1, at which visually discernible surface cracks manifest on a fully cured resin sample; (2) A pressure surge temperature, denoted as $T_3$ in Table 1, beyond which there is a significant escalation – a twofold increase within a 1 °C temperature increment – in the pressure within the vacuum chamber housing a fully cured resin sample. The baking temperatures selected for this study, represented as $T_0$ in Table 1, are deliberately set lower than both $T_2$ and $T_3$, thereby preserving both structural and vacuum-compatible integrity of the resin components. Furthermore, we undertake a quantitative evaluation of the volume shrinkage ratio pre- and post-thermal treatment. Our measurements

**Table 1**
Key properties of the three distinct photo-polymer resins in this study. Parameters denoted with the annotation (a) are directly derived from the technical specification sheets provided by the respective manufacturers. Those parameters accompanied by the annotation (b) represent values that have been systematically optimized in this work.

| Key parameters | ABS-like | Transparent | High-temperature |
|---|---|---|---|
| Model | Elegoo ABS-like | NOVA3D high transparency | Siraya Sculpt Ultra |
| Thermal resistance temperature $T_1^{(a)}$ | 80 °C | 50 °C | 220 °C |
| Structural failure temperature $T_2^{(b)}$ | 150 °C | 130 °C | 250 °C |
| Pressure surge temperature $T_3^{(b)}$ | 140 °C | 120 °C | 210 °C |
| Baking temperature in this work $T_0^{(b)}$ | 130 °C | 110 °C | 180 °C |
| Exposure time$^{(b)}$ | 7.5 s | 5 s | 2.5 s |
| Z layer thickness$^{(b)}$ | 50 μm | 50 μm | 50 μm |
| Post-cure duration$^{(b)}$ | 60 min | 5 min | 30 min |
| Hardness$^{(a)}$ | 75 D | 83 D | 99 D |
| Printing difficulty$^{(a)}$ | Easy | Easy | Hard |

indicate that the volume contraction at the designated baking temperatures remains below 1 %, substantiating the material stability under the specified thermal conditions.

We utilize an Elegoo Mars 2 Mono LCD MSLA 3D printer, a device characterized by an entry-level profile for this study. The printer has a standard lateral resolution of 50 μm and a variable Z-layer thickness ranging from 10 to 200 μm. Given its maximum build volume of 129 mm (L) × 80 mm (W) × 150 mm (H), this apparatus is well-suited to accommodate the fabrication of components for an expansive array of applications. The Elgoo Mars 2 is equipped with a 405 nm monochrome LCD, delivering a light intensity of approximately 2 mW/cm$^2$. To satisfy the stringent criteria requisite for UHV-compatibility, in conjunction with attaining both expedited printing speeds and dimensional accuracy, we optimize the exposure times and layer thickness for various resins, as detailed in Table 1. Our investigation reveals that reducing exposure time or increasing layer thickness detrimentally impact the curing process, leading to pronounced outgassing issues, particularly for components with larger thickness. Conversely, overexposure not only slows down the fabrication process but also incites lateral extrusion, thereby severely compromising machining precision.

Subsequent to the printing phase, a series of critical post-processing steps are undertaken to achieve UHV-compatibility. These encompass cleaning, post-curing, surface coating, and vacuum baking procedures. The methodological workflow and associated instrumentation are graphically elucidated in Fig. 1. Within this figure, circled numbers indicate the typical duration of the respective treatment steps, measured in hours.

**Firstly**, the freshly fabricated component is subjected to immersion in a 99 % isopropyl alcohol (IPA) solution for 10 mins, facilitated by agitation using a magnetic stirrer. This temporally optimized cleaning procedure serves to thoroughly remove residual uncured resin, thereby minimizing the likelihood of vacuum system contamination. Simultaneously, the process is optimized to avoid the detrimental effects of monomer extraction. Inadequate purging of uncured resin could lead to contamination and compromised vacuum performance. Conversely, excessive monomer extraction could undermine the mechanical robustness of the part, potentially manifesting as structural fissures attributed to material inhomogeneity [28–30].

**Secondly**, the component is subjected to an additional cleaning procedure utilizing an ultrasonic cleaner filled with Mean Green Super Strength cleaning and degreasing solution. The cleaning medium is maintained at a temperature of 50 °C for better cleaning efficacy. This ultrasonic cleaning phase is strategically deployed to eradicate residual uncured resin that may be embedded in micro-concavities or intricate geometric configurations present on the component's surface. Although this particular cleaning phase is frequently overlooked in conventional MSLA cleaning protocols, our findings underscore its criticality for the attainment of UHV-compatibility. The omission of this cleaning phase results in a substantial amplification of hydrocarbon outgassing; specifically, a three-to-five order of magnitude increase is observed during the subsequent vacuum baking procedures.

**Thirdly**, following the exhaustive cleaning processes, the surface of the fabricated component is scrutinized using an optical microscope. As depicted in the left panel of Fig. 2, the microscopy reveals a distinct pattern of micro-concavities characterized by a grid dimension approximately commensurate with the 50 μm resolution of the Elegoo Mars 2 Mono LCD screen. Utilizing imaging contrast as an empirical gauge, we roughly estimate the depth of these micro-concavities to be around 20 μm. It is posited that the increased surface area introduced by these micro-concavities could proportionally increase the rate of outgassing, thereby posing a potential obstacle to achieving UHV conditions. To address this issue, we build a simple surface coating apparatus for proof-of-principle experimentation, depicted in the central panel of Fig. 2. The fabricated component is mounted on a rapidly rotating stage, contrived using a CPU fan. Subsequently, a diluted solution comprising 5 % resin in IPA is incrementally dispensed onto the top surface of the spinning component from a syringe, at a rate of one droplet per 10-s interval. A 405 nm LED light is positioned above the component to facilitate the continuous polymerization of the applied resin solution. Owing to the rapid rotation and low viscosity of the resin solution, a majority of the liquid is dispelled via centrifugal force. However, the residual liquid entrapped within the micro-concavities is effectively polymerized. Our optimized protocol, requiring an approximate duration of 5 mins, demonstrably improves the surface quality, as corroborated by subsequent microscopic inspection presented in the right panel of Fig. 2. Based on a conservative estimation, this surface flatting process diminishes the total surface area of the component by approximately 50 %, which should concomitantly halve the outgassing rate. This significant reduction contributes to the component's UHV-compatibility.

**Fourthly**, the component is subjected to a post-curing procedure within an Elegoo Mercury curing station (405 nm, 25 W), with the aim of facilitating complete solidification. This is particularly pertinent for internal layers that may not have achieved full solidification during the printing stage. It should be noted that the penetrative capacities of the curing light constitute a critical parameter, the efficacy of which varies among different resin types. To systematically evaluate the penetrative capacities, we employ hollow resin cubes as test subjects. These cubes are fabricated with dimensions of 20 mm × 20 mm × 20 mm and a wall thickness of 1 mm. The hollow interior is subsequently filled with liquid resin through an aperture at the top with a diameter of 3 mm. A 30 mW, 405 nm laser light is then directed onto one side of the cube, allowing the light to penetrate the wall and facilitate the curing process for the

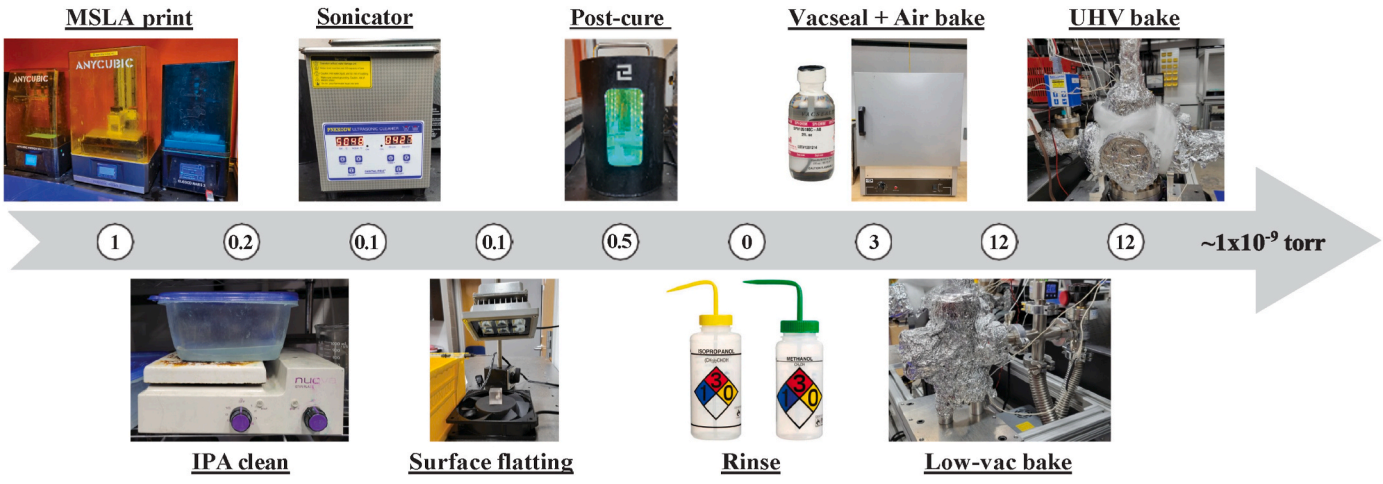



**Fig. 1.** A methodological workflow for fabrication of UHV-compatible components. The circled numbers indicate typical treatment durations, measured in hours.

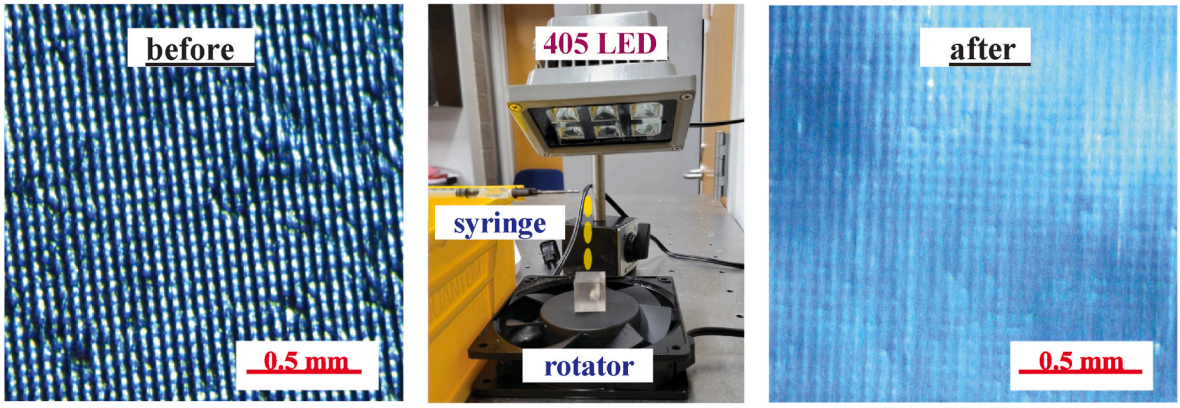



**Fig. 2.** Improvement of the surface quality through coating diluted resin. The left and right panels depict contrasting views of the surface conditions pre- and post-application of the surface coating. The central panel shows the custom-designed surface coating station.

internal liquid resin. Upon reaching a predetermined time interval, the laser is deactivated, the remaining liquid resin is drained, and the solidification depth of the internal resin is precisely measured. This procedure enables the mapping of a time-dependent profile for the penetration depth of the curing light, as illustrated in Fig. 3(b) and (c). Significantly, when utilizing transparent resin, it becomes feasible to directly observe the curing process through the wall of the cube, as shown in Fig. 3(a). This real-time monitoring of the curing procedure confirms that the penetration rate and depth are significantly better for transparent resins compared to ABS-like and high-temperature resin counterparts. Further experimentation indicates that heating the components to an approximate temperature of 60 °C serves to expedite the curing process, although this acceleration is observed only for high-temperature and transparent resins, not for ABS-like resin. To identify the optimal post-curing duration for each type of resin, we implement a linear power-dependent scaling approach. The requisite curing time for each resin is computed by scaling the observed curing rates in accordance with the light intensities of both the laser and the Elegoo Mercury curing station, as detailed in Table 1. Moreover, penetration depth metrics inform our recommendation that the maximum permissible thickness for components fabricated from ABS-like, high-temperature, and transparent resins should be constrained to 7 mm, 5 mm, and 30 mm, respectively. Importantly, these recommended values are twice the observed penetration depths detailed in Fig. 3, predicated on the

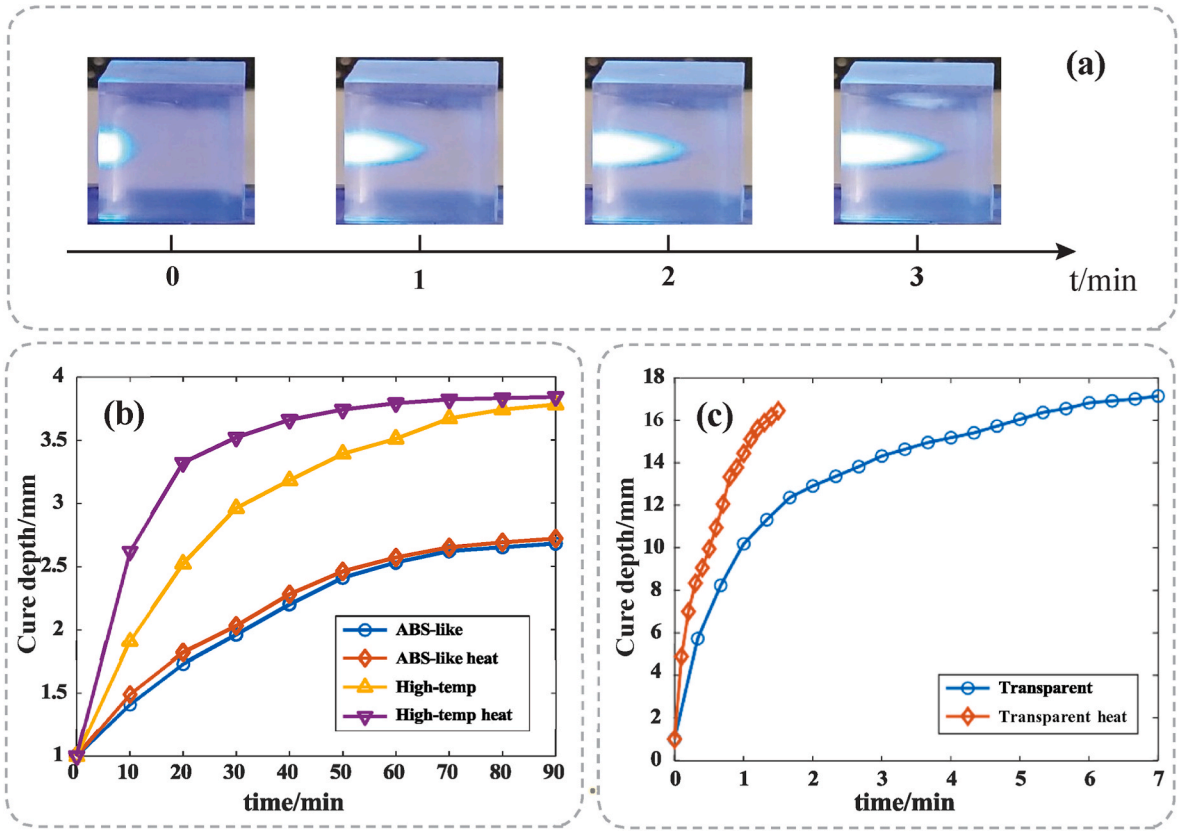



**Fig. 3.** Evaluation of Penetration Depth Across Different Resin Types. (a) Photographic representations capturing incremental changes in penetration depth as illuminated by a 405 nm, 30 mW laser at distinct time intervals. (b) Time-dependent penetration depth measurements of ABS-like and high-temperature resins. (c) Time-dependent penetration depth measurements of transparent resin, which has a much larger rate constant. Data collection ceases at a depth of 16 mm, a value comparable with the internal dimensions of the test hollow cube.

rationale that post-curing can be conducted from both sides of the component.

**Fifthly**, subsequent to the completion of the curing process, the component is subjected to a supplementary cleaning procedure. This involves a sequenced regimen of solvents – water, IPA, and methanol – aimed at the thorough removal of any residual contaminants. Following this cleaning phase, an optional step allows for the application of a Vacseal resin coating to the component's surface aimed for further reducing outgassing. In order to cure this Vacseal layer, the coated components are placed in an oven and exposed to ambient air for a duration extending over several hours. It is noteworthy that the manufacturer-recommended curing temperature for Vacseal resin is set at 200 °C for a 1-h interval. However, within the confines of our study, the upper thermal limits are constrained by the values denoted as the baking temperature $T_0$ in Table 1. The baking temperature for the high-temperature resin is commensurate with Vacseal's prescribed curing specifications. In contrast, the maximum allowed baking temperatures for ABS-like and transparent resins fall considerably below this benchmark, thereby introducing complications in achieving complete curing of the Vacseal coating.

**Sixthly**, once cleaned and optionally coated with Vacseal resin, the component is transferred into a low-vacuum chamber characterized by a base pressure of 0.7 mtorr. Within this environment, the component is subjected to a 12-h baking process at the baking temperature $T_0$, outlined in Table 1. The primary objective of this pre-baking stage is the volatilization and consequent removal of the majority of hydrocarbon molecules that may remain adhered to the component's surface. The omission of this critical step could have a deleterious contamination for the UHV conditions in subsequent stages. Residual hydrocarbon molecules possess the potential to contaminate not only the UHV chamber but also the ion gauge, thereby jeopardizing the integrity and reliability of the entire experimental apparatus. Therefore, this study introduces an additional quality control metric: only the components that successfully reach the targeted base pressure within the low-vacuum chamber are permitted to advance to the subsequent phases of experimentation. This quality control measure serves to ensure both the cleanliness of the components and their compatibility with UHV conditions.

**Finally**, following the low-vacuum pre-baking procedure, the component is promptly transferred to a UHV chamber equipped with a Pfeiffer TPH240 turbo pump, capable of maintaining a base pressure around $4 \times 10^{-10}$ torr. Within this environment, the component undergoes an additional 12-h baking process at temperatures consistent with those employed in the pre-baking stage. To facilitate monitoring and data acquisition of the chamber's pressure throughout this pivotal secondary baking process, we utilize a KJLC 354 Series hot filament ion gauge, characterized by an absolute measurement uncertainty of approximately 20 %. In order to substantiate the gauge's accuracy, cross-calibration exercises are conducted tri-monthly using another KJLC 354 Series ion gauge possessing a National Institute of Standards and Technology (NIST) traceable calibration certificate, in a separate UHV chamber. The observed pressure discrepancies between the two gauges consistently remain below 10 %, thereby affirming the reliability of our measurements. Additionally, a Residual Gas Analyzer (Microvision Plus RGA) is integrated into the system to scrutinize the species of outgassing. This inclusion affords a comprehensive spectral analysis of the internal gas constituents of the chamber, thereby providing invaluable insights into the effectiveness of each preparatory stage of the component. Such a methodology ensures compliance with the most stringent UHV-compatibility criteria, meeting or exceeding both scientific and industrial benchmarks.

## 3. Results and discussions

### 3.1. Hydrocarbon contamination

In the endeavor to attain optimal UHV conditions, we confront a number of significant challenges, among which hydrocarbon contamination emanating from inadequately cleaned and insufficiently post-cured components stands as a salient issue. Using the ABS-like resin as an example, we discover that circumventing key stages in the post-curing process – specifically durations falling short of a 10-min threshold – or neglecting the ultrasonic cleaning phase precipitates a mass loss on the order of approximately 4 % after the 120 °C baking. Such a reduction in mass significantly surpasses what can be accounted for by simple water evaporation and suggests the volatilization of either appreciable quantities of uncured resin or hydrocarbon molecules during the baking process. After the implementation of an optimized post-processing protocol, we successfully mitigated this mass loss to an inconsequential level, typically registering less than 0.1 %.

The implications of resin volatilization extend considerably beyond simple evaporation and warrant meticulous attention. Contrary to merely sublimating and being efficiently evacuated by the vacuum pump, the residual resin forms tenacious deposits on the internal walls of our experimental chamber. These accumulations not only present a complex cleaning dilemma but also exert a deleterious effect on the performance of our measurement instruments. Specifically, empirical observations reveal that the resin deposits critically impair the collecting electrode of the ion gauge, resulting in a dramatic five-fold reduction in its sensitivity. The eradication of such entrenched contamination necessitates not only an extended temporal investment but also significant manual labor. These complications underscore the imperative of rigorously optimizing our cleaning and post-curing protocols. Negligence in this regard compromises not only the accuracy of the UHV measurements but also raises questions concerning the long-term reliability and operational efficacy of the entire experimental setup.

### 3.2. Baking procedure

In most UHV experiments, the high-temperature baking phase frequently serves as the linchpin in accelerating the outgassing process. Detailed in Section 2, our experimental protocol incorporates a dual-stage baking strategy, incorporating both low-vacuum and high-vacuum environments. This stratified approach affords a heightened layer of protection, effectively mitigating risks associated with hydrocarbon contamination in the UHV system. For components that have been subjected to Vacseal resin coating, an initial air-baking phase is integrated into the protocol. The comprehensive sequence of these operations is graphically delineated in Fig. 4(a). Fig. 4(b) and (c) offer representative data sets that elucidate both temperature and pressure profiles during the respective low-vacuum and high-vacuum baking stages.

While the principal objective of this research endeavor is the attainment of a UHV environment, there exists an equally imperative need to optimize the baking procedure within a temporally feasible framework. This consideration gains heightened significance in the context of shared UHV systems, where protracted baking durations are often impracticable. To this end, we execute a comprehensive suite of time-dependent measurements across a spectrum of resin types, with an accentuated focus on high-temperature resin subjected to Vacseal coating, as delineated in Fig. 5. Consistent with our expectations, the empirical data corroborates that pressure levels exhibit a monotonically decreasing trend with the prolongation of the baking period. Among the investigated materials, the high-temperature resin coated with Vacseal distinguished itself as the most efficacious in engendering optimal vacuum conditions. A one-day baking period is sufficient to attain the UHV regime, registering pressure levels at approximately $1 \times 10^{-9}$ torr. For applications necessitating vacuum levels within the $1 \times 10^{-7}$ to $1 \times 10^{-8}$ torr range, a markedly truncated in-vacuum baking interval – spanning merely one to 2 h – is adequate. Furthermore, we undertake an evaluative scrutiny of the relative efficacy of low-vacuum and high-vacuum baking phases when utilizing high-temperature resin in conjunction with Vacseal coating. Preliminary results indicate that the

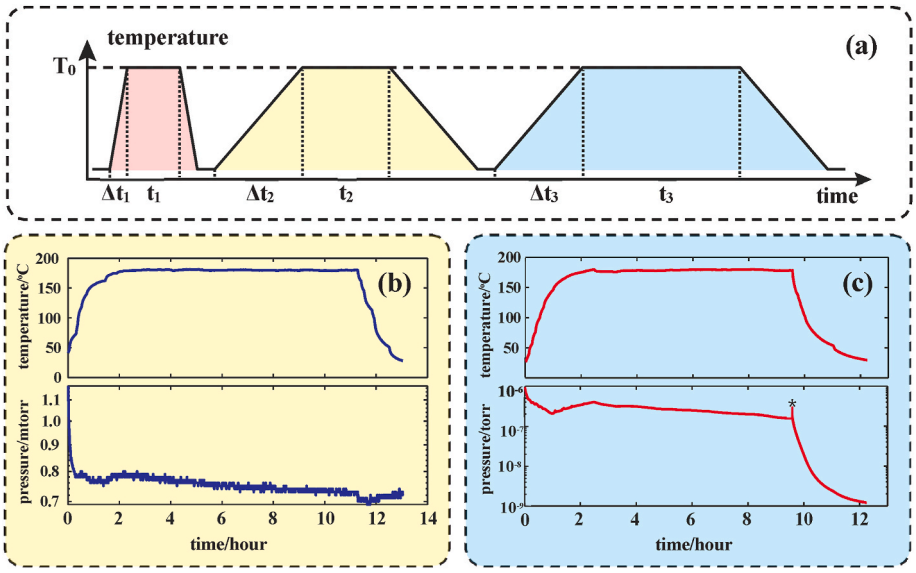



**Fig. 4.** Monitoring of temperatures and pressures throughout the baking procedure. (a) A schematic diagram shows an entire baking sequence, including the air bake (red), the low-vacuum bake (yellow), and the high-vacuum bake (blue). The air bake is exclusively applicable to resins treated with Vacseal coating. Typical time intervals for temperature increase and decrease for three types of baking are $\Delta t_1 = 1$ h, $\Delta t_1 = \Delta t_1 = 2$ h. The time intervals, $t_1$, $t_2$, $t_3$ corresponding to stable temperature plateaus during the baking process are systematically analyzed in Fig. 5. (b) Temperature and pressure changes during a typical low-vacuum baking procedure with high-temperature resin. (c) Temperature and pressure changes during a typical high-vacuum baking procedure with high temperature resin. The pressure surge annotated with an asterisk, is attribute to a 2-min ion gauge degas operation.

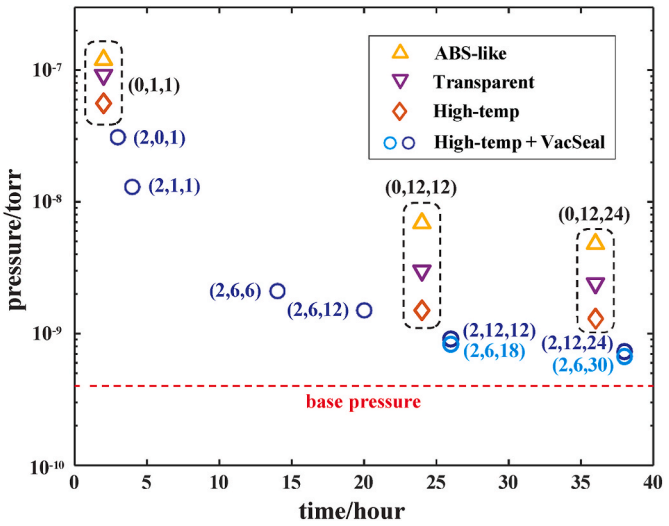



**Fig. 5.** Pressure measurements correlated with different baking durations. The triad of numerical values enclosed in parentheses signifies the respective time intervals allocated to air-baking, low-vacuum baking, and high-vacuum baking procedures, as elaborated in Fig. 4.

ultimate vacuum quality manifests a degree of resilience, remaining relatively invariant irrespective of the vacuum phase in which the baking is executed, contingent upon the constancy of the overall baking duration. This observation provides further flexibility in tailoring the baking process within the relatively simple low-vacuum chamber, thereby economizing on experimental resources while concurrently fulfilling specific experimental prerequisites.

### 3.3. Dimension analysis

To evaluate the scalability of our experimental methodology, especially with regard to the treatment of larger components, we undertake a systematic analysis of specimens representing five distinct geometrical configurations and fabricated from three different resins, as detailed in Table 2. All components selected for this investigation are subjected to a standardized thermal treatment process, including a 12-h pre-baking stage immediately succeeded by a 12-h UHV baking stage. The temperatures employed for the thermal treatment of each resin type are contingent upon their material properties and are explicitly documented in Table 1. For components that have been treated with a Vacseal coating, an additional air-baking procedure is conducted for a duration of 2 h, employing the maximal temperature tolerances specific to each resin type.

Our empirical findings reveal a consistent trend: the final vacuum pressures are predominantly influenced by two morphological attributes of the components – surface area and wall thickness. Although the absolute pressure values manifest variation across the three different resin types, the rates of pressure increase with respect to dimensional alternations display remarkable similarity. Notably, components that have been treated with a Vacseal coating exhibit a notable departure from this overarching trend. More specifically, components constructed from high-temperature resin manifest a considerable reduction in their final pressure values. In contrast, components fabricated from ABS-like and transparent resins show an increment in the final pressure values. This

**Table 2**
Summary of the final pressures achieved across a range of resin components following the completion of an approximately 30-h post-printing process.

| Dimension | ABS-like | Transparent | High-temp | Notes |
|---|---|---|---|---|
| 20 mm × 20 mm × 2 mm | $8.6 \times 10^{-10}$ | $9.2 \times 10^{-10}$ | $4.8 \times 10^{-10}$ | |
| 80 mm × 20 mm × 2 mm | $1.1 \times 10^{-9}$ | $1.2 \times 10^{-9}$ | $4.6 \times 10^{-10}$ | |
| 80 mm × 20 mm × 4 mm | $2.8 \times 10^{-9}$ | $1.6 \times 10^{-9}$ | $7.2 \times 10^{-10}$ | |
| 20 mm × 20 mm × 10 mm | $6.9 \times 10^{-9}$ | $3.0 \times 10^{-9}$ | $1.5 \times 10^{-9}$ | |
| 20 mm × 20 mm × 10 mm | $1.1 \times 10^{-8}$ | $7.4 \times 10^{-9}$ | $9.1 \times 10^{-10}$ | Vacseal coating |
| 80 mm × 20 mm × 10 mm | not test | not test | $1.2 \times 10^{-9}$ | Vacseal coating |

discrepancy is primarily attributable to the degree of the completeness achieved in the curing process of the Vacseal coating during the 2-h air-bake stage. In the case of high-temperature resin, the Vacseal coating reaches full maturation at a temperature of 180 °C, whereas the corresponding coatings for ABS-like and transparent resins do not achieve full curing at a temperature of 120 °C. By leveraging the application of Vacseal coating, we have successfully attained a final vacuum pressure of $1.2 \times 10^{-9}$ torr for the largest component fabricated using high-temperature resin.

### 3.4. Residual gas analysis

To obtain more information of the molecular species contributing to outgassing, we employ a Residual Gas Analyzer (RGA). As the UHV chamber and resin components undergo temperature increase, molecules prone to outgassing, including water and hydrocarbon molecules, are evaporated from the surface of the components. Subsequent to their release, these volatilized molecules are ionized or dissociated via high-energy electrons within the RGA that is directly interfaced with the UHV chamber. Following this ionization or dissociation process, the resultant charged molecular fragments are channeled into a quadrupole mass spectrometer for mass-selective detection and analysis. Because the species detected are fragmentary constituents of the outgassed molecules, a single molecular species may yield multiple fragments with distinct masses, each detectable through disparate channels in the RGA. Within the scope of our experiment, approximately eight salient fragments have been identified, as delineated in Fig. 6(a). To ascertain whether multiple fragments originate from a single molecular species, we conduct temperature-dependent measurements, thereby generating eight distinct pressure curves as functions of temperature, as portrayed in Fig. 6(b). Normalizing these curves with respect to the pressure value at 120 °C facilitates the categorization of these fragments into two discrete groups, each displaying a uniform temperature-pressure dependency. The fragments with mass-to-charge ratios of 17 and 18 amu are unambiguously identified as water molecules. Consequently, the remaining group of fragments could be attributed to hydrocarbon molecules. Based on this preliminary investigation, our current analytical capabilities are insufficient to ascertain the detailed chemical composition of the hydrocarbon molecules.

Employing molecular fragments with 18 and 45 amu as representative indicators for water and hydrocarbon molecules respectively, we investigate the temperature-dependent degassing profiles across three types of resins, as well as high-temperature resin with a Vacseal coating. Fig. 6(c) elucidates that the high-temperature resin is characterized by a substantially reduced emission of water molecules during elevated-temperature baking relative to the other resins. This observed discrepancy may originate either from inherent material variations or from the different temperatures applied during the pre-baking stage, during which the high-temperature resin is exposed to a significantly higher temperature. Notably, the high-temperature resin treated with Vacseal coating exhibits a markedly distinct pattern of water molecule release. The low levels of detected water vapor pressure indicate the effectiveness of the Vacseal coating as an optimal surface sealant, thereby inhibiting the absorption of atmospheric water or its subsequent desorption under vacuum conditions. Furthermore, the declining trajectory of the graph signifies the expeditious evacuation of any remnant water vapor within the UHV chamber by the turbo pump upon temperature escalation, typically transpiring within a 2-h timespan.

In a manner consistent with the observed behavior for water molecules, Fig. 6(d) shows similar outgassing trends for hydrocarbon molecules. Notably, the high-temperature resin is characterized by a diminished emission of hydrocarbon molecules, even when subjected to high baking temperatures, in contrast to the other two resin types. Most significantly, the high-temperature resin treated with Vacseal coating essentially eradicates the phenomenon of hydrocarbon molecule outgassing, even during the entire UHV baking process. These observations

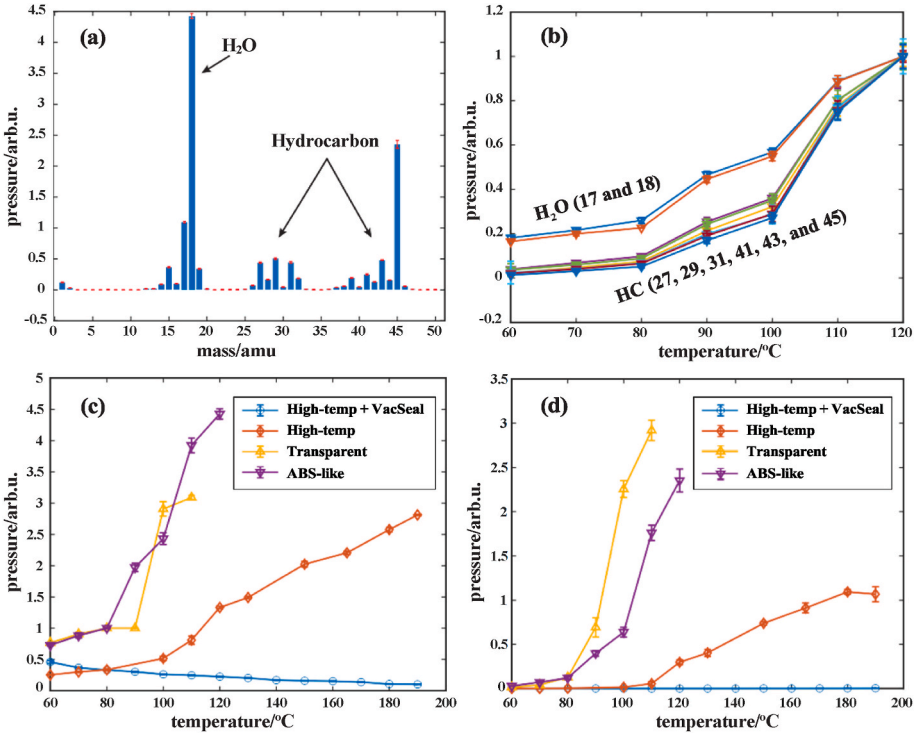



**Fig. 6.** RGA measurement results of outgassing process in the UHV chamber. (a) The RGA detects molecular fragments with mass-to-charge ratios ranging from 1 to 50 amu during the UHV baking process at a temperature of 120 °C, utilizing ABS-like resin as the test material. (b) Molecular fragments originating from both water and hydrocarbon molecules exhibit distinctive temperature-dependent vapor pressure profiles. The partial pressure for each fragment is normalized to the value at 120 °C. (c) The vapor pressure of water molecules is monitored as a function of temperature during the UHV baking phase. (d) The vapor pressure of hydrogen molecules is monitored as a function of temperature during the UHV baking phase.

furnish a qualitative rationale for the high-temperature resin's superior performance in attaining optimized vacuum levels within equivalent baking durations. Additionally, the incorporation of a Vacseal coating onto the high-temperature resin serves as an effective countermeasure against the potential contamination of the UHV environment by hydrocarbon molecules, a subject that has been emphasized in Section 3.1.

## 4. Conclusions and outlook

In this study, we have conclusively demonstrated the feasibility of achieving UHV-compatibility – measured at approximately $1 \times 10^{-9}$ torr – via the strategic application of economically efficient AM techniques. Employing high-temperature photo-polymer resins, we develop a post-treatment protocol specifically designed to facilitate UHV-compatibility. This research endeavor responds to an emergent and critical need for methodologies that not only permit the achievement of UHV conditions, but also enable the expeditious prototyping of geometrically intricate components. Through the successful integration of the rapidity and adaptability intrinsic to AM with the exacting specifications mandated by UHV environments, we introduce a novel approach that allows for the cost-effective and efficient fabrication of small, yet complexly structured, components. This synergetic approach substantially truncates the time required for proof-of-concept evaluations, thereby imparting a marked augmentation to the efficiency of research and development endeavors.

In contemplating future prospects, we adopt an optimistic outlook regarding the feasibility of elevating attainable vacuum levels by an additional order of magnitude to $1 \times 10^{-10}$ torr or beyond. Reaching this landmark would substantially broaden the scope of practical applications, particularly in the research of ultracold atomic and molecular physics, surface chemistry, and material science. To actualize this ambitious objective, we identify multiple avenues for optimization: **Firstly**, the extant limitations imposed by the curing process during the AM process constitute a significant constraint on the base pressure that can be achieved. Although post-curing methodologies yield some enhancements, constraints related to penetration depth—especially in the context of high-temperature and ABS-like resins—impose a cap on the maximum component thickness. The forthcoming advent of next-generation resin printers, furnished with Digital Mirror Devices (DMD) and more highly focused light sources, holds the promise of improving in-printing cure quality, thereby positively influencing vacuum performance. **Secondly**, our RGA measurements, particularly those concerning high-temperature resin treated with Vacseal coating, have yielded encouraging preliminary results. These findings suggest that the Vacseal layer could act as an efficacious barrier against the majority of hydrocarbon outgassing. The deployment of an enhanced vacuum system capable of achieving an order-of-magnitude improvement in base pressure would facilitate a more granular chemical analysis of any extant residual gases. Moreover, the utilization of a two-stage vacuum system could alleviate contamination risks manifesting during the sample loading phase, thereby elevating the vacuum environment's purity. **Lastly**, the extension of the vacuum bake duration within the UHV chamber, potentially up to one week, may offer a pathway to further mitigate ultimate pressure levels. Such a protracted baking process would furnish additional time for the efflux of residual gases from the material, thereby stretching the current limitations on vacuum performance by an additional factor of two.

In addition to striving for enhanced vacuum performance, we are also intrigued by another research avenue that seeks to hybrid UHV-compatible AM with traditional subtractive manufacturing techniques. AM excels in its ability to create complex geometries quickly and cost-effectively, thereby expanding the range of feasible designs. On the other hand, traditional subtractive manufacturing methods bring unique material properties to the table, such as higher tensile strength, superior thermal conductivity, and improved surface finish. By integrating these two manufacturing paradigms into a hybridized approach, it is possible to leverage the strengths of each to create components that not only meet the intricate geometric specifications often required in advanced research settings but also possess enhanced material properties that are essential for high-performance applications. This hybridization could impact diverse sectors, from advanced materials science to precision engineering, by providing a pathway to build components with a unique combination of complexity and material integrity.

## CRediT authorship contribution statement

**Jiaqi Li:** Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Validation, Writing – original draft, Writing – review & editing. **Thomas Mcpartland:** Investigation, Methodology, Writing – review & editing. **Bernardo Gutierrez:** Validation, Writing – review & editing. **Jan Pedersen:** Conceptualization, Project administration, Supervision, Funding acquisition, Writing – review & editing. **Yan Zhou:** Writing – review & editing, Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Validation, Visualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## Acknowledgement

We thank Prof. Kwong at UNLV for lending the RGA. Jiaqi Li thanks the UNLV TTDGRA support. Bernardo Gutierrez thanks the NSF EPSCoR Fellowship. This work is financially supported by UNLV Faculty Opportunity Awards.

## References

[1] H. Kodama, Automatic method for fabricating a three-dimensional plastic model with photo-hardening polymer, Rev. Sci. Instrum. 52 (1981) 1770–1773, https://doi.org/10.1063/1.1136492.
[2] K.V. Wong, A. Hernandez, A review of additive manufacturing, ISRN Mech. Eng. 2012 (2012), 208760, https://doi.org/10.5402/2012/208760.
[3] H. Bikas, P. Stavropoulos, G. Chryssolouris, Additive manufacturing methods and modeling approaches: a critical review, Int. J. Adv. Manuf. Technol. 83 (2016) 389–405, https://doi.org/10.1007/s00170-015-7576-2.
[4] A.P. Fagundes, J.O.D.B. Lira, N. Padoin, C. Soares, H.G. Riella, Additive manufacturing of functional devices for environmental applications: a review, J. Environ. Chem. Eng. 10 (2022), https://doi.org/10.1016/j.jece.2022.108049.
[5] Z. Huang, G. Shao, L. Li, Micro/nano functional devices fabricated by additive manufacturing, Prog. Mater. Sci. 131 (2023), 101020, https://doi.org/10.1016/j.pmatsci.2022.101020.
[6] A. Wiberg, J. Persson, J. Ölvander, Design for additive manufacturing – a review of available design methods and software, Rapid Prototyp. J. 25 (2019) 1080–1094, https://doi.org/10.1108/RPJ-10-2018-0262.
[7] D.M. Nieto, D.M. Sánchez, Design for additive manufacturing: tool review and a case study, Appl. Sci. 11 (2021) 1–13, https://doi.org/10.3390/app11041571.
[8] B. Regassa Hunde, A. Debebe Woldeyohannes, Future prospects of computer-aided design (CAD) – a review from the perspective of artificial intelligence (AI), extended reality, and 3D printing, Results Eng. 14 (2022), 100478, https://doi.org/10.1016/j.rineng.2022.100478.
[9] B. Zhang, A. Goel, O. Ghalsasi, S. Anand, CAD-based design and pre-processing tools for additive manufacturing, J. Manuf. Syst. 52 (2019) 227–241, https://doi.org/10.1016/j.jmsy.2019.03.005.
[10] A. Vyatskikh, S. Delalande, A. Kudo, X. Zhang, C.M. Portela, J.R. Greer, Additive manufacturing of 3D nano-architected metals, Nat. Commun. 9 (2018), https://doi.org/10.1038/s41467-018-03071-9.
[11] Y. Yang, G. Wang, H. Liang, C. Gao, S. Peng, L. Shen, C. Shuai, Additive manufacturing of bone scaffolds, Int. J. Bioprinting 5 (2019) 1–25, https://doi.org/10.18063/IJB.v5i1.148.
[12] F. Navah, M.-E. Lamarche-Gagnon, F. Ilinca, Thermofluid topology optimization for cooling channel design, Appl. Therm. Eng. 236 (2023), 121317, https://doi.org/10.1016/j.applthermaleng.2023.121317 arXiv:2302.04745.

[13] I. Paoletti, Mass customization with additive manufacturing: new perspectives for multi performative building components in architecture, Procedia Eng. 180 (2017) 1150–1159, https://doi.org/10.1016/j.proeng.2017.04.275.
[14] T. Srivatsan, T. Sudarshan, Additive Manufacturing Innovations, Advances, and Applications, CRC Press, 2016.
[15] S. Jung, L.B. Kara, Z. Nie, T.W. Simpson, K.S. Whitefoot, Is additive manufacturing an environmentally and economically preferred alternative for mass production? Environ. Sci. Technol. 57 (2023) 6373–6386, https://doi.org/10.1021/acs.est.2c04927.
[16] H.A. Colorado, E.I. Velásquez, S.N. Monteiro, Sustainability of additive manufacturing: the circular economy of materials and environmental perspectives, J. Mater. Res. Technol. 9 (2020) 8221–8234, https://doi.org/10.1016/j.jmrt.2020.04.062.
[17] https://www.ge.com/additive/.
[18] S.W. Paek, S. Balasubramanian, D. Stupples, Composites additive manufacturing for space applications: a review, Materials 15 (2022) 4709, https://doi.org/10.3390/ma15134709.
[19] A. Charles, A. Hofer, A. Elkaseer, S.G. Scholz, Additive manufacturing in the automotive industry and the potential for driving the green and electric transition, Sustain. Des. Manuf. 262 (2022) 339–346, https://doi.org/10.1007/978-981-16-6128-0_32.
[20] Q. Yan, H. Dong, J. Su, J. Han, B. Song, Q. Wei, Y. Shi, A review of 3d printing technology for medical applications, Engineering 4 (2018) 729–742, https://doi.org/10.1016/j.eng.2018.07.021.
[21] P.R. Johnson, P.M. Copeland, A.O. Ayodele, E.N. Tarekegn, S.J. Bromley, W. R. Harrell, C.E. Sosolik, J.P. Marler, In-vacuum performance of a 3d-printed ion deflector, Vacuum 172 (2020), 109061, https://doi.org/10.1016/j.vacuum.2019.109061.
[22] A.P. Povilus, C.J. Wurden, Z. Vendeiro, M. Baquero-Ruiz, J. Fajans, Vacuum compatibility of 3d-printed materials, J. Vac. Sci. Technol. A: Vacuum, Surfaces, and Films 32 (2014), 033001, https://doi.org/10.1116/1.4873556.
[23] N. Bihari, I.T.S. Heikkinen, G. Marin, C. Ekstrum, P.J. Mayville, S. Oberloier, H. Savin, M. Karppinen, J.M. Pearce, Vacuum outgassing characteristics of unpigmented 3D printed polymers coated with atomic layer deposited alumina, J. Vac. Sci. Technol. A 38 (2020), 053204, https://doi.org/10.1116/6.0000178.
[24] I.T. Heikkinen, G. Marin, N. Bihari, C. Ekstrum, P.J. Mayville, Y. Fei, Y.H. Hu, M. Karppinen, H. Savin, J.M. Pearce, Atomic layer deposited aluminum oxide mitigates outgassing from fused filament fabrication–based 3-D printed components, Surf. Coating. Technol. 386 (2020), 125459, https://doi.org/10.1016/j.surfcoat.2020.125459.
[25] N. Cooper, L.A. Coles, S. Everton, R.P. Campion, S. Madkhaly, C. Morley, W. Evans, R. Saint, P. Krüger, F. Oručević, C. Tuck, R.D. Wildman, T.M. Fromhold, L. Hackermueller, Additively manufactured ultra-high vacuum chamber below $10^{-10}$ mbar, arXiv:1903 (2019), 07708, https://doi.org/10.48550/arXiv.1903.07708.
[26] J. Vovrosh, G. Voulazeris, P.G. Petrov, J. Zou, Y. Gaber, L. Benn, D. Woolger, M. M. Attallah, V. Boyer, K. Bongs, M. Holynski, Additive manufacturing of magnetic shielding and ultra-high vacuum flange for cold atom sensors, Sci. Rep. 8 (2018) 1–10, https://doi.org/10.1038/s41598-018-20352-x.
[27] T.D. Ngo, A. Kashani, G. Imbalzano, K.T. Nguyen, D. Hui, Additive manufacturing (3D printing): a review of materials, methods, applications and challenges, Compos. B Eng. 143 (2018) 172–196, https://doi.org/10.1016/j.compositesb.2018.02.012.
[28] A. Bardelcik, S. Yang, F. Alderson, A. Gadsden, The effect of wash treatment on the mechanical properties and energy absorption potential of a 3D printed polymethyl methacrylate (PMMA), Mater. Today Commun. 26 (2021), 101728, https://doi.org/10.1016/j.mtcomm.2020.101728.
[29] A.C. Uzcategui, A. Muralidharan, V.L. Ferguson, S.J. Bryant, R.R. McLeod, Understanding and improving mechanical properties in 3D printed parts using a dual-cure acrylate-based resin for Stereolithography, Adv. Eng. Mater. 20 (2018) 1–10, https://doi.org/10.1002/adem.201800876.
[30] N. Vaidya, O. Solgaard, 3D printed optics with nanometer scale surface roughness, Microsyst. Nanoeng. 4 (2018), https://doi.org/10.1038/s41378-018-0015-4.